

BOȘȚINĂ GEORGE EDUARD (Eduard Bostina)

CS Student | Linux Kernel & Systems Programming

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Third-year Computer Science student at Politehnica University of Bucharest (GPA 9.4/10) contributing to the mainline Linux kernel through Google Summer of Code with The Linux Foundation. Lead architect of a 64-bit AArch64 OS kernel running on real hardware with SMP and virtual memory. **Seeking a Summer 2027 systems/embedded engineering internship; open to relocation across the EU.**

PROFESSIONAL EXPERIENCE

The Linux Foundation | Google Summer of Code May 2026 - Present
Contributor, Linux Kernel | Remote

- Converted 50 legacy text-based Device Tree bindings to YAML dt-schema, most from Texas Instruments, enabling automated hardware-description validation across the ARM/OMAP platform tree.
- Patches reviewed and accepted for mainline via linux-next, iterating with subsystem maintainers on the devicetree and OMAP lists through the upstream LKML review workflow; further series in review.
- Verifiable upstream work: lore.kernel.org/all/?q=egbostina@gmail.com

EDUCATION

Politehnica University of Bucharest Oct 2024 - Jul 2028 (expected)
Faculty of Automatic Control and Computer Science
B.Sc. in Computer Science & Engineering (in progress) **GPA: 9.4 / 10**

PROJECTS

- Perspicua** C, AArch64, Assembly 2025 - Present | github.com/perspicua/perspicua
Lead architect and primary contributor on a 4-person 64-bit AArch64 kernel for the Raspberry Pi 4B, written from scratch in C and assembly (~31k LOC). **Boots on real hardware to a userspace shell:** SMP across 4 cores, virtual memory with copy-on-write, 37 POSIX-style syscalls (fork/exec, signals, pipes, mmap), a VFS with FAT32 root, and page/block caches. 19 in-kernel test suites, KASAN, lockdep, and an in-kernel debugger.
- Aegis** C++17, libsodium, zlib 2025 | github.com/edybostina/aegis
Cross-platform CLI for authenticated file encryption (Linux/macOS/Windows) using AEAD (XChaCha20-Poly1305). Passphrase and keyfile modes, optional compression, recursive directory processing, documented security design; 5 tagged releases with CI via GitHub Actions.
- Matrix-lib** C++17, SIMD, Multithreading 2025 | github.com/edybostina/matrix-lib-cpp
Header-only C++ linear-algebra library: 50+ operations, LU/QR decomposition, and eigenvalues. Optimized with SIMD (AVX2/NEON) and automatic multithreading for **4-8x speedups**, plus an optional BLAS backend.
- Rasterizer** C++, SDL2 2025 | github.com/edybostina/rasterizer
Real-time software 3D renderer with no GPU. Full pipeline: vertex transformation, view-frustum clipping, and scan-line rasterization to a pixel buffer. Multithreaded; loads .obj models with texture mapping.

TRAINING, AWARDS & CERTIFICATIONS

Google ARM Summer School | Embedded Linux Summer 2025
Selected participant | Politehnica Bucharest | One-week intensive

- Implemented a complete ARM boot chain (TF-A, U-Boot, kernel) and a custom Buildroot Linux distribution (kernel, rootfs, initramfs) for the NXP i.MX93 (Cortex-A55), including TrustZone/TEE secure-world configuration and custom device drivers.

Awards & Certifications

- Gold Medalist (Top 1%), National Olympiad in Applied Informatics, AcadNet 2024**
- Global Finalist (6th worldwide, 2nd in Romania), JetBrains Student Quiz 2025
- NVIDIA Certificate: Building AI Agents with Multimodal Models (2025)

TECHNICAL SKILLS

Languages	C/C++, Assembly (AArch64, NASM), Python, Bash/Shell, C#, Java
Systems & Tools	Linux, Buildroot, U-Boot, QEMU, GDB, CMake, Git, Docker, GitHub Actions (CI/CD), SDL2, CUDA
Expertise	Device drivers, kernel modules, Device Tree, cross-compilation, POSIX, concurrency, memory management, embedded Linux, systems programming
Spoken	Romanian (native), English (fluent)